

FINAL PROJECT END REPORT

COMP.SE.140-2024-2025

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Github URL: <https://github.com/duyle95/devops-docker-compose.git>

Gitlab URL: <https://compse140.devops-gitlab.rd.tuni.fi/mtc382/devops-final-project.git>

Branch name: project

Optional features: None

Deployment:

- Local deployment only, no external server deployment
- Can be deployed locally with the docker-compose commands

CI/CD pipeline:

- CI can be run locally by executing the script “test.sh”.
- Register a Gitlab runner with Shell executor, you should be able to run CI/CD pipeline in the latest commit and the whole project will be deployed in your local machine.
- Successful deployment will produce in these containers:

	devops-final-p	Running (4/4)	0.02%	4 minutes ago
	api-gateway 7a36e5bfc26c	devops-final-project- Running	0% 8197:8197 ↗	4 minutes ago
	frontend-1 218e28763f8c	devops-final-project- Running	0% 8198:80 ↗	4 minutes ago
	golang-serv f803f8fd9861	devops-final-project- Running	0.02%	4 minutes ago
	node-servic 54817b5aedd1	devops-final-project- Running	0%	4 minutes ago

Testing:

- Manual testing with curl commands should work for all endpoints
- Testing framework: Jest, there are very basic integration tests and will be runned automatically if Gitlab runner is registered.

Example logs of passing test:

Duy Le / devops-final-project / Jobs / #10672

Search visible log output

```
158 Container ci-node-service-1 Starting
159 Container ci-frontend-1 Starting
160 Container ci-api-gateway-1 Starting
161 Container ci-api-gateway-1 Started
162 Container ci-node-service-1 Started
163 Container ci-frontend-1 Started
164 Container ci-golang-service-1 Started
165 time="2025-01-26T22:47:06+02:00" level=warning msg="Found orphan containers ([ci-frontend-1 ci-golang-service-1 ci-node-service-1 ci-api-gateway-1]) for th
is project. If you removed or renamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up."
166 Container ci-tests-1 Creating
167 Container ci-tests-1 Created
168 Container ci-tests-1 Starting
169 Container ci-tests-1 Started
170 TEST_EXIT_CODE: '0'
171 Tests Passed
172 Container ci-api-gateway-1 Killing
173 Container ci-frontend-1 Killing
174 Container ci-node-service-1 Killing
175 Container ci-golang-service-1 Killing
176 Container ci-golang-service-1 Killed
177 Container ci-node-service-1 Killed
178 Container ci-frontend-1 Killed
179 Container ci-api-gateway-1 Killed
180 Going to remove ci-tests-1, ci-frontend-1, ci-golang-service-1, ci-node-service-1, ci-api-gateway-1
181 Container ci-api-gateway-1 Removing
182 Container ci-frontend-1 Removing
183 Container ci-tests-1 Removing
184 Container ci-golang-service-1 Removing
185 Container ci-node-service-1 Removing
186 Container ci-golang-service-1 Removed
187 Container ci-node-service-1 Removed
188 Container ci-api-gateway-1 Removed
189 Container ci-frontend-1 Removed
190 Container ci-tests-1 Removed
191 Cleaning up project directory and file based variables
192 Job succeeded
```

Duration: 43 seconds
Finished: 3 minutes ago
Queued: 3 seconds
Timeout: 1h (from project)
Runner: #195 (Q_FqmqzPP)

Commit 49526a08

Tests should work and ci/cd should be successful

Pipeline #3193 Passed for project

test

Related jobs

→ test

Example logs of failing test:

Duy Le / devops-final-project / Jobs / #10669

Search visible log output

```
158 Container ci-api-gateway-1 Started
159 Container ci-golang-service-1 Started
160 Container ci-frontend-1 Started
161 Container ci-node-service-1 Started
162 time="2025-01-26T22:45:08+02:00" level=warning msg="Found orphan containers ([ci-node-service-1 ci-frontend-1 ci-api-gateway-1 ci-golang-service-1]) for th
is project. If you removed or renamed this service in your compose file, you can run this command with the --remove-orphans flag to clean it up."
163 Container ci-tests-1 Recreate
164 Container ci-tests-1 Recreated
165 Container ci-tests-1 Starting
166 Container ci-tests-1 Started
167 TEST_EXIT_CODE: '1'
168 Tests Failed - Exit Code: 1
169 Container ci-golang-service-1 Killing
170 Container ci-frontend-1 Killing
171 Container ci-api-gateway-1 Killing
172 Container ci-node-service-1 Killing
173 Container ci-golang-service-1 Killed
174 Container ci-node-service-1 Killed
175 Container ci-api-gateway-1 Killed
176 Container ci-frontend-1 Killed
177 Going to remove ci-tests-1, ci-node-service-1, ci-frontend-1, ci-api-gateway-1, ci-golang-service-1
178 Container ci-golang-service-1 Removing
179 Container ci-api-gateway-1 Removing
180 Container ci-node-service-1 Removing
181 Container ci-tests-1 Removing
182 Container ci-frontend-1 Removing
183 Container ci-api-gateway-1 Removed
184 Container ci-golang-service-1 Removed
185 Container ci-frontend-1 Removed
186 Container ci-node-service-1 Removed
187 Container ci-tests-1 Removed
188 Cleaning up project directory and file based variables
189 ERROR: Job failed: exit status 1
```

Duration: 35 seconds
Finished: 3 minutes ago
Queued: 3 seconds
Timeout: 1h (from project)
Runner: #195 (Q_FqmqzPP)

Commit af1cd154

Give permission to execute test.sh

Pipeline #3192 Failed for project

test

Related jobs

→ test

Reflection:

- Main learnings:
 - I have never set up a CI pipeline, so this is a good learning experience overall
- Effort used:
 - 20-30 hours so far, I would want to deploy to GCP but didn't got the time.